Telescopes and binoculars

We see things because light from them produces an image on the retina at the back of each eye. Faraway things produce only a small image on the retina. A telescope or a pair of binoculars produces a magnified image, which takes up more space on the retina.

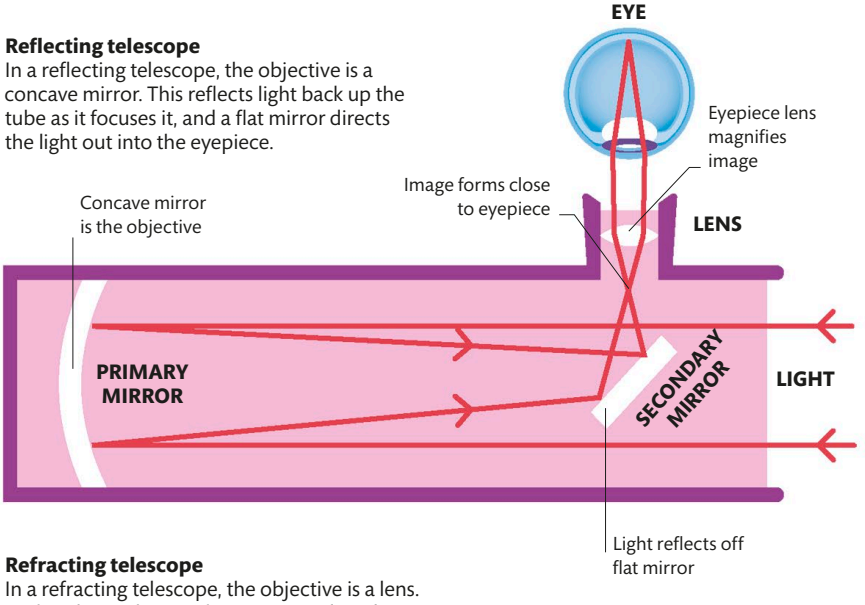
WHAT DO THE TWO NUMBERS ON BINOCULARS MEAN?

On binoculars marked 10x50, 10 refers to the magnification and 50 to the diameter of the two objective lenses, in millimeters.

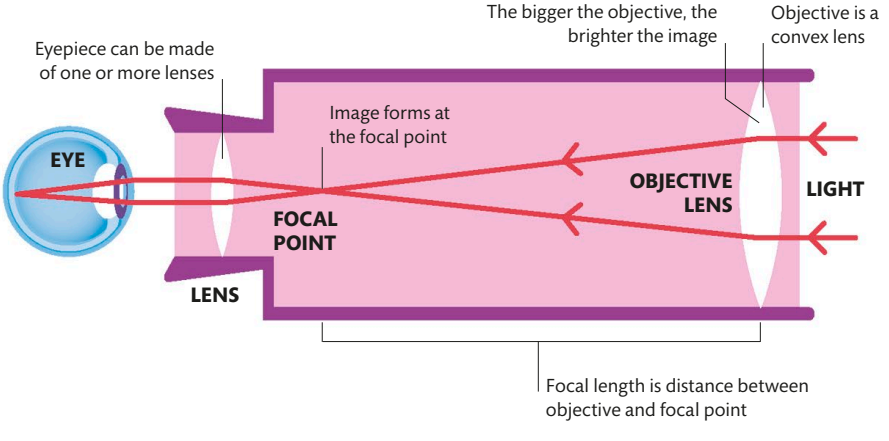
Telescopes

In a telescope, a lens or mirror, called the objective, focuses light from a distant object. This makes an image of the object inside the tube. The eyepiece lens magnifies the image. The longer the focal length (distance between the lens or mirror and the point where the rays meet) of the objective, the larger the image in the tube. The shorter the focal length of the eyepiece lens, the larger the image appears in your eye.

Reflecting telescope
In a reflecting telescope, the objective is a concave mirror. This reflects light back up the tube as it focuses it, and a flat mirror directs the light out into the eyepiece.

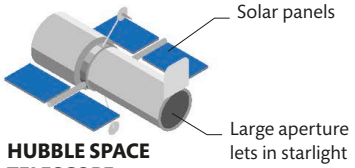


Refracting telescope
In a refracting telescope, the objective is a lens. With only two lenses, the image produced is upside down, so some refracting telescopes contain more lenses to correct this.



SPACE TELESCOPES

The atmosphere absorbs some of the light coming from distant planets, stars, and galaxies, and its turbulent motion reduces image quality. Space telescopes do not suffer these problems. Images are captured digitally and beamed back to Earth.



HUBBLE SPACE TELESCOPE